## A3.2 Basic read and write transactions

This section defines the basic mechanisms for AXI protocol transactions. The basic mechanisms are:

- the *Handshake process*
- the *Channel signaling requirements* on page A3-38.

### A3.2.1 Handshake process

All five transaction channels use the same **VALID**/**READY** handshake process to transfer address, data, and control information. This two-way flow control mechanism means both the master and slave can control the rate at which the information moves between master and slave. The *source* generates the **VALID** signal to indicate when the address, data or control information is available. The *destination* generates the **READY** signal to indicate that it can accept the information. Transfer occurs only when *both* the **VALID** and **READY** signals are HIGH.

On master and slave interfaces there must be no combinatorial paths between input and output signals.

Figure A3-2 to Figure A3-4 on page A3-38 show examples of the handshake process.

In Figure A3-2, the source presents the address, data or control information after T1 and asserts the **VALID** signal. The destination asserts the **READY** signal after T2, and the source must keep its information stable until the transfer occurs at T3, when this assertion is recognized.

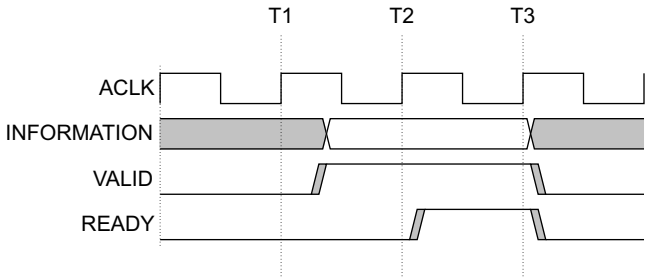



**Figure A3-2 VALID before READY handshake**

A source is not permitted to wait until **READY** is asserted before asserting **VALID**.

Once **VALID** is asserted it must remain asserted until the handshake occurs, at a rising clock edge at which **VALID** and **READY** are both asserted.

In Figure A3-3, the destination asserts **READY**, after T1, before the address, data or control information is valid, indicating that it can accept the information. The source presents the information, and asserts **VALID**, after T2, and the transfer occurs at T3, when this assertion is recognized. In this case, transfer occurs in a single cycle.

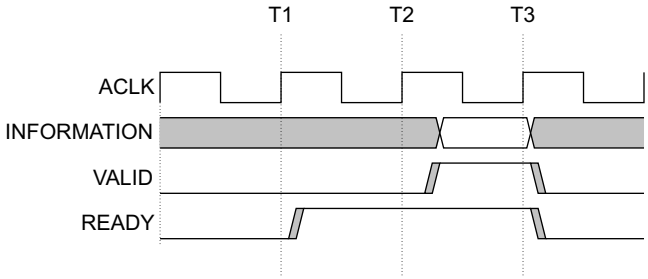



**Figure A3-3 READY before VALID handshake**

A destination is permitted to wait for **VALID** to be asserted before asserting the corresponding **READY**.

If **READY** is asserted, it is permitted to deassert **READY** before **VALID** is asserted.


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In Figure A3-4, both the source and destination happen to indicate, after T1, that they can transfer the address, data or control information. In this case the transfer occurs at the rising clock edge when the assertion of both **VALID** and **READY** can be recognized. This means the transfer occurs at T2.

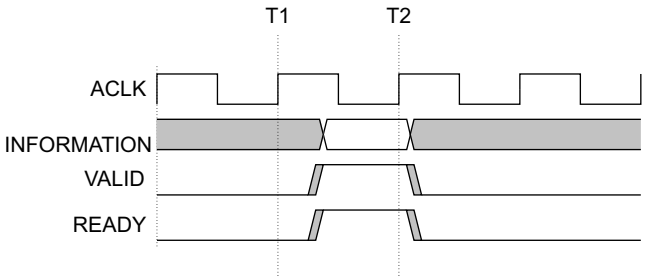



**Figure A3-4 VALID with READY handshake**

The individual AXI protocol channel handshake mechanisms are described in *Channel signaling requirements*.

## A3.2.2 Channel signaling requirements

The following sections define the handshake signals and the handshake rules for each channel:

- *Channel handshake signals*
- *Write address channel*
- *Write data channel* on page A3-39
- *Write response channel* on page A3-39
- *Read address channel* on page A3-39
- *Read data channel* on page A3-39.

### Channel handshake signals

Each channel has its own **VALID**/**READY** handshake signal pair. Table A3-1 shows the signals for each channel.

**Table A3-1 Transaction channel handshake pairs**

| Transaction channel | Handshake pair |
|---|---|
| Write address channel | **AWVALID**, **AWREADY** |
| Write data channel | **WVALID**, **WREADY** |
| Write response channel | **BVALID**, **BREADY** |
| Read address channel | **ARVALID**, **ARREADY** |
| Read data channel | **RVALID**, **RREADY** |

### Write address channel

The master can assert the **AWVALID** signal only when it drives valid address and control information. **When asserted, AWVALID** must remain asserted until the rising clock edge after the slave asserts **AWREADY**.

The default state of **AWREADY** can be either HIGH or LOW. This specification recommends a default state of HIGH. When **AWREADY** is HIGH the slave must be able to accept any valid address that is presented to it.

**Note**

This specification does not recommend a default **AWREADY** state of LOW, because it forces the transfer to take at least two cycles, one to assert **AWVALID** and another to assert **AWREADY**.


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### Write data channel

During a write burst, the master can assert the **WVALID** signal only when it drives valid write data. When asserted, **WVALID** must remain asserted until the rising clock edge after the slave asserts **WREADY**.

The default state of **WREADY** can be HIGH, but only if the slave can always accept write data in a single cycle.

The master must assert the **WLAST** signal while it is driving the final write transfer in the burst.

### Write response channel

The slave can assert the **BVALID** signal only when it drives a valid write response. When asserted, **BVALID** must remain asserted until the rising clock edge after the master asserts **BREADY**.

The default state of **BREADY** can be HIGH, but only if the master can always accept a write response in a single cycle.

### Read address channel

The master can assert the **ARVALID** signal only when it drives valid address and control information. When asserted, **ARVALID** must remain asserted until the rising clock edge after the slave asserts the **ARREADY** signal.

The default state of **ARREADY** can be either HIGH or LOW. This specification recommends a default state of HIGH. If **ARREADY** is HIGH then the slave must be able to accept any valid address that is presented to it.

**Note**

This specification does not recommend a default **ARREADY** value of LOW, because it forces the transfer to take at least two cycles, one to assert **ARVALID** and another to assert **ARREADY**.

### Read data channel

The slave can assert the **RVALID** signal only when it drives valid read data. When asserted, **RVALID** must remain asserted until the rising clock edge after the master asserts **RREADY**. Even if a slave has only one source of read data, it must assert the **RVALID** signal only in response to a request for data.

The master interface uses the **RREADY** signal to indicate that it accepts the data. The default state of **RREADY** can be HIGH, but only if the master is able to accept read data immediately, whenever it starts a read transaction.

The slave must assert the **RLAST** signal when it is driving the final read transfer in the burst.


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